

Part 3: The eyeball, its vessels and nerves, and clinical cases

140 questions in 5 sections. Answers and short explanations are at the end.

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How to use these questions. Each question has four statements and **only one** is correct, as in the real exam. Some questions ask for the statement that is FALSE (marked EXCEPT); read the stem carefully. Pick one answer, then check the key. The questions follow the lecture slides; where a slide and the textbooks disagree, the explanation says so.

Eyeball I: coats, aqueous humour, refractive media, retina

1. Which is NOT a coat of the eyeball?

- ☐ A. Neural (retina)
- ☐ B. Fibrous (sclera and cornea)
- ☐ C. Synovial
- ☐ D. Vascular (uvea)

2. Tenon's capsule (fascia bulbi):

- ☐ A. Is part of the retina
- ☐ B. Lines the inner surface of the sclera
- ☐ C. Envelops the eye from the optic nerve to the corneal limbus
- ☐ D. Covers the cornea

3. The cornea: all of the following are correct EXCEPT:

- ☐ A. Forms five sixths of the outer coat
- ☐ B. Is completely avascular
- ☐ C. Is highly sensitive (V1)
- ☐ D. Is nourished by the lacrimal fluid outside and aqueous humour inside

4. The sclera:

- ☐ A. Is more sensitive than the cornea
- ☐ B. Is the main refractive surface of the eye
- ☐ C. Forms one sixth of the outer coat
- ☐ D. Provides insertion for the extraocular muscles

5. The episclera: all of the following are correct EXCEPT:

- ☐ A. Attaches to Tenon's capsule
- ☐ B. Is part of the retina
- ☐ C. Is loose fibroelastic tissue
- ☐ D. Is the outermost layer of the sclera

6. Scleral icterus is associated with:

- ☐ A. Glaucoma
- ☐ B. Hepatitis and other liver diseases
- ☐ C. Conjunctivitis
- ☐ D. Cataract

7. Scleritis:

- ☐ A. Is painless
- ☐ B. Involves only the conjunctiva
- ☐ C. Is often associated with systemic disease such as rheumatoid arthritis
- ☐ D. Is a disease of the lens

8. Which is NOT part of the uvea?

- ☐ A. Retina
- ☐ B. Ciliary body
- ☐ C. Iris
- ☐ D. Choroid

9. The ciliary body:

- ☐ A. Lies behind the ora serrata
- ☐ B. Forms the blind spot
- ☐ C. Connects the choroid to the iris
- ☐ D. Belongs to the fibrous coat

10. Miosis is produced by:

- ☐ A. The sphincter pupillae (parasympathetic)
- ☐ B. Darkness
- ☐ C. The dilator pupillae
- ☐ D. Sympathetic stimulation

11. The dilator pupillae is supplied by:

- ☐ A. Parasympathetic fibres from the ciliary ganglion
- ☐ B. Sympathetic fibres
- ☐ C. CN III
- ☐ D. The facial nerve

12. Aqueous humour is secreted:

- ☐ A. By the cornea into the anterior chamber
- ☐ B. By the retina into the vitreous
- ☐ C. By the iris into the anterior chamber
- ☐ D. By the ciliary processes into the posterior chamber

13. Compared with plasma, aqueous humour contains less:

- ☐ A. Ascorbate
- ☐ B. Proteins
- ☐ C. Lactate
- ☐ D. Chloride

14. Functions of aqueous humour: all of the following are correct EXCEPT:

- ☐ A. Supplies oxygen and nutrients to the cornea and lens
- ☐ B. Absorbs light to prevent scattering
- ☐ C. Maintains intraocular pressure
- ☐ D. Removes waste and debris

15. Most aqueous humour is formed by:

- ☐ A. Filtration through the cornea
- ☐ B. Active secretion across the ciliary epithelium
- ☐ C. Secretion by the lens
- ☐ D. Secretion by the iris

16. The path of aqueous humour:

- ☐ A. Ciliary body → posterior chamber → pupil → anterior chamber → drainage at the angle
- ☐ B. Lens → anterior chamber
- ☐ C. Anterior chamber → pupil → posterior chamber
- ☐ D. Vitreous → posterior chamber → pupil

17. The trabecular (conventional) outflow route:

- ☐ A. Drains less than 10% of the aqueous
- ☐ B. Passes through the ciliary muscle
- ☐ C. Is pressure independent
- ☐ D. Passes through Schlemm's canal

18. The uveoscleral outflow route:

- ☐ A. Is pressure sensitive
- ☐ B. Passes through the trabecular meshwork only
- ☐ C. Passes through Schlemm's canal
- ☐ D. Is pressure independent

19. In ocular inflammation:

- ☐ A. Aqueous production stops
- ☐ B. Prostaglandins block uveoscleral outflow
- ☐ C. Intraocular pressure always falls
- ☐ D. The trabecular meshwork can become clogged

20. Intraocular pressure is measured with:

- ☐ A. A tonometer
- ☐ B. A retinoscope
- ☐ C. An ophthalmoscope
- ☐ D. A perimeter

21. The refractive media in order are:

- ☐ A. Cornea, lens, aqueous, vitreous
- ☐ B. Aqueous, cornea, vitreous, lens
- ☐ C. Lens, cornea, aqueous, vitreous
- ☐ D. Cornea, aqueous humour, lens, vitreous humour

22. Where does the maximum refraction of light occur?

- ☐ A. Lens
- ☐ B. Aqueous humour
- ☐ C. Cornea
- ☐ D. Vitreous

23. The lens:

- ☐ A. Lies in front of the iris
- ☐ B. Is suspended by zonular fibres
- ☐ C. Is biconcave
- ☐ D. Is supplied by the central retinal artery

24. In accommodation for near vision:

- ☐ A. The ciliary muscle contracts and the zonules relax
- ☐ B. The ciliary muscle relaxes and the zonules tighten
- ☐ C. Sympathetic fibres drive the ciliary muscle
- ☐ D. The lens becomes thinner

25. At rest (far vision): all of the following are correct EXCEPT:

- ☐ A. The pupil is maximally constricted
- ☐ B. The zonules are taut
- ☐ C. The ciliary muscle is relaxed
- ☐ D. The lens is flattened

26. The subcapsular epithelium of the lens lies:

- ☐ A. On both surfaces
- ☐ B. On the posterior surface only
- ☐ C. On the anterior surface only
- ☐ D. Nowhere; it disappears in adults

27. Presbyopia is due to:

- ☐ A. A corneal ulcer
- ☐ B. Optic nerve damage
- ☐ C. Detachment of the retina
- ☐ D. Loss of elasticity of the lens

28. Cataract is:

- ☐ A. Clouding of the lens
- ☐ B. Inflammation of the uvea
- ☐ C. Swelling of the optic disc
- ☐ D. Damage to the optic nerve

29. The vitreous body:

- ☐ A. Transmits light and helps hold the retina in place
- ☐ B. Is secreted by the lens
- ☐ C. Fills the anterior chamber
- ☐ D. Is continuously drained through Schlemm's canal

30. The neural layer of the retina ends anteriorly at:

- ☐ A. The limbus
- ☐ B. The pupillary margin
- ☐ C. The ora serrata
- ☐ D. The optic disc

31. The pigmented layer of the retina:

- ☐ A. Forms the optic nerve
- ☐ B. Is several cells thick
- ☐ C. Absorbs light to reduce scattering
- ☐ D. Contains the photoreceptors

32. The optic disc:

- ☐ A. Lies in the centre of the macula
- ☐ B. Has the highest density of cones
- ☐ C. Contains only rods
- ☐ D. Is the blind spot

33. The fovea:

- ☐ A. Lies in the centre of the macula
- ☐ B. Is the blind spot
- ☐ C. Contains only rods
- ☐ D. Is where the optic nerve leaves the eye

34. Rods:

- ☐ A. Give colour vision
- ☐ B. Contain rhodopsin and serve dim-light vision
- ☐ C. Number about 6 million per retina
- ☐ D. Are concentrated in the fovea

35. Cones:

- ☐ A. Contain rhodopsin
- ☐ B. Number about 120 million per retina
- ☐ C. Serve scotopic vision
- ☐ D. Give colour vision and high acuity

36. Which cells generate action potentials in the retina?

- ☐ A. Pigment cells
- ☐ B. Ganglion cells
- ☐ C. Bipolar cells
- ☐ D. Photoreceptors

37. Which cells convert light into nerve signals?

- ☐ A. Ganglion cells
- ☐ B. Horizontal cells
- ☐ C. Bipolar cells
- ☐ D. Photoreceptors (rods and cones)

38. The optic nerve is formed by:

- ☐ A. Axons of amacrine cells
- ☐ B. Axons of bipolar cells
- ☐ C. Axons of ganglion cells
- ☐ D. Processes of rods

Eyeball II: extraocular muscles, retina and the visual pathway

39. The inferior oblique arises from:

- ☐ A. The body of the sphenoid
- ☐ B. The anterior floor of the orbit
- ☐ C. The common tendinous ring
- ☐ D. The lesser wing of the sphenoid

40. The tendon of the superior oblique passes through:

- ☐ A. The superior orbital fissure
- ☐ B. The common tendinous ring
- ☐ C. The lacrimal fossa
- ☐ D. The trochlea

41. The inferior oblique is supplied by:

- ☐ A. CN VI
- ☐ B. CN IV
- ☐ C. The superior branch of CN III
- ☐ D. The inferior branch of CN III

42. Which is an action of the superior rectus?

- ☐ A. Extorsion
- ☐ B. Abduction
- ☐ C. Depression
- ☐ D. Intorsion

43. Which is an action of the inferior rectus?

- ☐ A. Extorsion
- ☐ B. Elevation
- ☐ C. Abduction
- ☐ D. Intorsion

44. Which is an action of the superior oblique?

- ☐ A. Extorsion
- ☐ B. Elevation
- ☐ C. Depression
- ☐ D. Adduction

45. Which is an action of the inferior oblique?

- ☐ A. Adduction
- ☐ B. Depression
- ☐ C. Intorsion
- ☐ D. Extorsion

46. Which pair of muscles produces intorsion?

- ☐ A. Inferior oblique and inferior rectus
- ☐ B. Inferior oblique and superior rectus
- ☐ C. Superior oblique and inferior rectus
- ☐ D. Superior oblique and superior rectus

47. Which pair of muscles produces extorsion?

- ☐ A. Superior oblique and inferior rectus
- ☐ B. Inferior oblique and superior rectus
- ☐ C. Superior oblique and superior rectus
- ☐ D. Inferior oblique and inferior rectus

48. Which pair of muscles elevates the eye?

- ☐ A. Inferior rectus and superior oblique
- ☐ B. Superior rectus and superior oblique
- ☐ C. Inferior rectus and inferior oblique
- ☐ D. Superior rectus and inferior oblique

49. Which pair of muscles depresses the eye?

- ☐ A. Inferior rectus and superior oblique
- ☐ B. Superior rectus and superior oblique
- ☐ C. Superior rectus and inferior oblique
- ☐ D. Inferior rectus and inferior oblique

50. Which muscle does NOT abduct the eye?

- ☐ A. Lateral rectus
- ☐ B. Inferior oblique
- ☐ C. Superior oblique
- ☐ D. Medial rectus

51. Which muscle does NOT adduct the eye?

- ☐ A. Superior rectus
- ☐ B. Lateral rectus
- ☐ C. Inferior rectus
- ☐ D. Medial rectus

52. The medial rectus is supplied by:

- ☐ A. The inferior branch of CN III
- ☐ B. The superior branch of CN III
- ☐ C. CN VI
- ☐ D. CN IV

53. The non-visual retina:

- ☐ A. Covers the ciliary body and the back of the iris
- ☐ B. Contains the fovea
- ☐ C. Is sensitive to light
- ☐ D. Contains the optic disc

54. In the retina, light travels:

- ☐ A. Directly to ganglion cells first
- ☐ B. In the same direction as the signals
- ☐ C. Only through the pigmented layer first
- ☐ D. In the opposite direction to the flow of neuronal signals

55. The visual pathway in order:

- ☐ A. Retina, optic nerve, medial geniculate, cortex
- ☐ B. Retina, optic nerve, chiasm, optic tract, lateral geniculate nucleus, optic radiation, visual cortex
- ☐ C. Retina, optic tract, chiasm, optic nerve, cortex
- ☐ D. Retina, chiasm, optic nerve, radiation, tract

56. At the optic chiasm:

- ☐ A. All fibres cross
- ☐ B. Fibres from the nasal half of each retina cross
- ☐ C. Fibres from the temporal half of each retina cross
- ☐ D. No fibres cross

57. The lateral geniculate nucleus is:

- ☐ A. Part of the occipital lobe
- ☐ B. A relay for hearing
- ☐ C. Part of the midbrain tectum
- ☐ D. A visual relay centre in the thalamus

58. The primary visual cortex lies in:

- ☐ A. The frontal lobe
- ☐ B. The temporal lobe
- ☐ C. The parietal lobe
- ☐ D. The occipital lobe

59. A complete cut of the left optic nerve causes:

- ☐ A. Total loss of vision in the left eye
- ☐ B. Right homonymous hemianopia
- ☐ C. Bitemporal hemianopia
- ☐ D. No visual loss

60. A lesion of the optic chiasm causes:

- ☐ A. Blindness in one eye
- ☐ B. Homonymous hemianopia
- ☐ C. Loss of the temporal visual fields of both eyes (bitemporal hemianopia)
- ☐ D. Loss of the nasal fields

61. A lesion of the left optic tract causes:

- ☐ A. Loss of vision in the left eye
- ☐ B. Identical loss of the right half of the visual field in both eyes
- ☐ C. Left homonymous hemianopia
- ☐ D. Bitemporal hemianopia

62. Partial damage to the left optic nerve affecting temporal retinal fibres causes loss of:

- ☐ A. The nasal visual field of the left eye
- ☐ B. The temporal visual field of the left eye
- ☐ C. The right visual field of both eyes
- ☐ D. Both temporal fields

63. Which step is NOT part of the pupillary light reflex pathway?

- ☐ A. Ciliary ganglion to the dilator pupillae
- ☐ B. Pretectum to the Edinger-Westphal nuclei
- ☐ C. Projections from the retina to the pretectum
- ☐ D. CN III to the ciliary ganglion

64. Why does light in one eye constrict both pupils?

- ☐ A. Both optic nerves cross completely
- ☐ B. The pretectal nuclei project to both Edinger-Westphal nuclei
- ☐ C. The ciliary ganglion is bilateral
- ☐ D. CN VII is involved

Vascular supply and applied anatomy of the eye

65. The ophthalmic artery enters the orbit through:

- ☐ A. The optic canal
- ☐ B. The inferior orbital fissure
- ☐ C. The superior orbital fissure
- ☐ D. The foramen rotundum

66. Which is NOT an orbital branch of the ophthalmic artery?

- ☐ A. Supraorbital
- ☐ B. Anterior and posterior ethmoidal
- ☐ C. Lacrimal
- ☐ D. Maxillary

67. Which is NOT an ocular branch of the ophthalmic artery?

- ☐ A. Central retinal artery
- ☐ B. Lacrimal artery
- ☐ C. Long ciliary arteries
- ☐ D. Short ciliary arteries

68. The central retinal artery:

- ☐ A. Supplies the choroid
- ☐ B. Enters the optic nerve inside the eyeball
- ☐ C. Supplies the retina
- ☐ D. Is a branch of the facial artery

69. The short ciliary arteries supply:

- ☐ A. The choroid
- ☐ B. The eyelids
- ☐ C. The lacrimal gland
- ☐ D. The inner layers of the retina

70. The lacrimal artery supplies:

- ☐ A. The ethmoidal cells
- ☐ B. The retina
- ☐ C. The lacrimal gland and lateral eyelids
- ☐ D. The superior oblique

71. The supraorbital artery supplies:

- ☐ A. The forehead and eyebrow
- ☐ B. The retina
- ☐ C. The maxillary sinus
- ☐ D. The inferior rectus

72. The anterior ethmoidal artery supplies:

- ☐ A. Only the posterior ethmoidal cells
- ☐ B. The anterior and middle ethmoidal cells and the frontal sinus
- ☐ C. The maxillary sinus
- ☐ D. The retina

73. Which muscle is NOT supplied by the inferior muscular branch of the ophthalmic artery?

- ☐ A. Inferior oblique
- ☐ B. Medial rectus
- ☐ C. Inferior rectus
- ☐ D. Superior oblique

74. The superior ophthalmic vein drains into:

- ☐ A. The sigmoid sinus
- ☐ B. The cavernous sinus
- ☐ C. The external jugular vein
- ☐ D. The superior sagittal sinus

75. The vortex veins drain:

- ☐ A. The lacrimal gland
- ☐ B. The uveal tract
- ☐ C. The eyelids
- ☐ D. The retina, into the facial vein

76. Hypertropia is:

- ☐ A. Downward turning of the eye
- ☐ B. Outward turning of the eye
- ☐ C. Upward turning of the eye
- ☐ D. Inward turning of the eye

77. Strabismus:

- ☐ A. Most commonly turns one eye outward
- ☐ B. Means the eyes do not point in the same direction
- ☐ C. Is clouding of the lens
- ☐ D. Never results from nerve damage

78. Nystagmus: all of the following are correct EXCEPT:

- ☐ A. May be horizontal, vertical or rotary
- ☐ B. Is caused by a cloudy lens
- ☐ C. Is repetitive uncontrolled eye movement
- ☐ D. Usually results from brain injury

79. In myopia:

- ☐ A. The image falls behind the retina
- ☐ B. The image falls in front of the retina
- ☐ C. Near objects are blurred
- ☐ D. The eyeball is usually too short

80. In hyperopia:

- ☐ A. Distant objects are blurred and near ones clear
- ☐ B. The image is focused behind the retina
- ☐ C. The eyeball is usually too long
- ☐ D. The image falls in front of the retina

81. Which orbital wall is most commonly affected by a blowout fracture?

- ☐ A. Medial wall
- ☐ B. Lateral wall
- ☐ C. Roof
- ☐ D. Floor

82. In a medial wall blowout fracture, which muscle may herniate into the ethmoid air cells?

- ☐ A. Lateral rectus
- ☐ B. Superior oblique
- ☐ C. Inferior rectus
- ☐ D. Medial rectus

83. Corneal abrasion: all of the following are correct EXCEPT:

- ☐ A. Is usually traumatic (fingernail, grit, contact lens)
- ☐ B. Is a disease of the lens
- ☐ C. Usually heals within a few days
- ☐ D. Can lead to infection or scarring if not healing

84. Ptosis can result from:

- ☐ A. Paralysis of orbicularis oculi
- ☐ B. Loss of sympathetic supply to the superior tarsal muscle
- ☐ C. Lateral rectus palsy
- ☐ D. Trochlear nerve palsy

85. Which is NOT a feature of facial nerve palsy?

- ☐ A. Difficulty closing the eye
- ☐ B. Drooping corner of the mouth
- ☐ C. Loss of corneal sensation
- ☐ D. Inability to raise the eyebrow on one side

86. A chalazion is:

- ☐ A. An infection of the lacrimal gland
- ☐ B. A blocked meibomian gland within the lid
- ☐ C. A clouding of the lens
- ☐ D. A blocked gland at the lash line (stye)

87. Bacterial conjunctivitis typically causes:

- ☐ A. A cloudy lens
- ☐ B. Inflammation of the uvea
- ☐ C. A mucopurulent discharge
- ☐ D. Swelling of the optic disc

88. Uveitis affects:

- ☐ A. The outer conjunctiva
- ☐ B. The cornea only
- ☐ C. The middle (vascular) layer of the eye
- ☐ D. The lens only

89. Papilloedema is:

- ☐ A. Swelling of the optic disc from raised intracranial pressure
- ☐ B. Swelling of the eyelid
- ☐ C. Inflammation of the conjunctiva
- ☐ D. Clouding of the lens

90. Glaucoma:

- ☐ A. Is easily cured completely
- ☐ B. Is clouding of the lens
- ☐ C. Never involves raised eye pressure
- ☐ D. Damages the optic nerve and causes visual field loss

91. Retinal detachment often presents with:

- ☐ A. Flashes and floaters
- ☐ B. Yellow sclera
- ☐ C. A painful red eye with discharge
- ☐ D. A painless lump in the lid

Motor and sensory innervation of the eye

92. Which cranial nerve is purely sensory?

- ☐ A. IX
- ☐ B. VII
- ☐ C. VIII
- ☐ D. V

93. Which cranial nerve is purely motor?

- ☐ A. V
- ☐ B. IX
- ☐ C. VII
- ☐ D. IV

94. Which cranial nerve carries parasympathetic fibres?

- ☐ A. IV
- ☐ B. VI
- ☐ C. III
- ☐ D. V

95. The optic nerve: all of the following are correct EXCEPT:

- ☐ A. Is wrapped in meninges
- ☐ B. Behaves like a tract of the brain
- ☐ C. Is surrounded by CSF in the subarachnoid space
- ☐ D. Has no meningeal covering

96. Third-order neurons of the visual pathway run from:

- ☐ A. The retina to the chiasm
- ☐ B. The lateral geniculate body to the cortex
- ☐ C. The bipolar cells to the ganglion cells
- ☐ D. The rods and cones to the bipolar cells

97. The primary visual cortex is Brodmann area:

- ☐ A. 17
- ☐ B. 41
- ☐ C. 4
- ☐ D. 22

98. The trochlear nucleus lies:

- ☐ A. In the medulla
- ☐ B. In the midbrain, at the level of the superior colliculus
- ☐ C. In the midbrain, at the level of the inferior colliculus
- ☐ D. In the pons

99. A left midbrain infarct involving the trochlear nucleus causes:

- ☐ A. Bilateral ptosis
- ☐ B. A right CN VI palsy
- ☐ C. Left facial palsy
- ☐ D. A left CN IV palsy (per the lecture)

100. Interneurons of the abducens nucleus cross to:

- ☐ A. The contralateral facial nucleus
- ☐ B. The contralateral oculomotor nucleus
- ☐ C. The Edinger-Westphal nucleus
- ☐ D. The ipsilateral trochlear nucleus

101. The medial longitudinal fasciculus:

- ☐ A. Carries the pupillary light reflex
- ☐ B. Links the retina to the lateral geniculate body
- ☐ C. Links CN III, IV, VI and VIII for conjugate eye movements
- ☐ D. Carries the corneal reflex

102. A right pontine infarct damages the right abducens nucleus. The paralysed muscle is:

- ☐ A. Right lateral rectus
- ☐ B. Left lateral rectus
- ☐ C. Right medial rectus
- ☐ D. Left medial rectus

103. During accommodation for near vision:

- ☐ A. The ciliary muscle relaxes
- ☐ B. The lens becomes more convex
- ☐ C. Dioptric power decreases
- ☐ D. The suspensory ligament becomes taut

104. When focusing from near to far:

- ☐ A. The lens becomes flatter
- ☐ B. The pupils constrict
- ☐ C. The ciliary muscle contracts
- ☐ D. Dioptric power increases

105. Which is NOT part of the near response (triad)?

- ☐ A. Pupillary constriction
- ☐ B. Divergence of the eyes
- ☐ C. Accommodation of the lens
- ☐ D. Convergence of the eyes

106. Which is part of the afferent limb of the accommodation reflex?

- ☐ A. The Edinger-Westphal nucleus
- ☐ B. The visual cortex
- ☐ C. The ciliary ganglion
- ☐ D. The short ciliary nerves

107. The efferent limb of the accommodation reflex starts in:

- ☐ A. The Edinger-Westphal nucleus
- ☐ B. The pretectal nucleus
- ☐ C. The facial nucleus
- ☐ D. The superior cervical ganglion

108. Cones have:

- ☐ A. Low acuity and no colour
- ☐ B. High acuity but low sensitivity to light (high threshold)
- ☐ C. High sensitivity to dim light
- ☐ D. Rhodopsin as their pigment

109. Rods have:

- ☐ A. Colour vision
- ☐ B. A high threshold
- ☐ C. High acuity
- ☐ D. Great sensitivity to light but low acuity

110. For the pupillary light reflex, optic tract fibres project to:

- ☐ A. The lateral geniculate body and then the cortex
- ☐ B. The superior cervical ganglion
- ☐ C. The facial nucleus
- ☐ D. The pretectal nuclei

111. Which is NOT part of the efferent pathway of the pupillary light reflex?

- ☐ A. Ciliary ganglion and short ciliary nerves
- ☐ B. Dilator pupillae
- ☐ C. Edinger-Westphal nucleus
- ☐ D. Oculomotor nerve

112. In darkness:

- ☐ A. Only one pupil dilates
- ☐ B. The light reflex is enhanced
- ☐ C. Both pupils dilate
- ☐ D. Both pupils constrict

113. The consensual light reflex is:

- ☐ A. Constriction of the illuminated eye only
- ☐ B. Constriction of the pupil of the eye that is not illuminated
- ☐ C. Dilation of the other pupil
- ☐ D. Carried by the facial nerve

114. The efferent limb of the corneal reflex is:

- ☐ A. The trochlear nerve
- ☐ B. The ophthalmic nerve (V1)
- ☐ C. The oculomotor nerve
- ☐ D. The facial nerve to orbicularis oculi

115. An absent corneal reflex may indicate all of the following EXCEPT:

- ☐ A. Deep coma
- ☐ B. Stroke
- ☐ C. Myopia
- ☐ D. Trigeminal or facial nerve lesion

116. Orbicularis oculi is supplied by:

- ☐ A. The temporal and zygomatic branches of the facial nerve
- ☐ B. The buccal and marginal mandibular branches of the facial nerve
- ☐ C. The oculomotor nerve
- ☐ D. The trigeminal nerve

117. Which is NOT part of the parasympathetic pathway to the lacrimal gland?

- ☐ A. Facial nerve → greater (superficial) petrosal nerve
- ☐ B. Ciliary ganglion
- ☐ C. Pterygoid canal
- ☐ D. Zygomatic and lacrimal nerves to the gland

118. In the ciliary ganglion:

- ☐ A. Only the parasympathetic fibres synapse
- ☐ B. The sympathetic fibres synapse
- ☐ C. All fibres synapse
- ☐ D. The sensory fibres synapse

119. The parasympathetic root of the ciliary ganglion arrives via:

- ☐ A. The facial nerve
- ☐ B. The nasociliary nerve
- ☐ C. The motor root from the inferior division of CN III
- ☐ D. The internal carotid plexus

120. The oculomotor nerve:

- ☐ A. Leaves the skull through the inferior orbital fissure
- ☐ B. Supplies the masseter
- ☐ C. Has its nucleus in the pons
- ☐ D. Supplies levator palpebrae superioris

Clinical cases (mixed)

121. A patient's right eye is 'down and out', with a dilated pupil and complete ptosis. The lesion involves:

- ☐ A. Right abducens nerve
- ☐ B. Right facial nerve
- ☐ C. Right oculomotor nerve
- ☐ D. Right trochlear nerve

122. A patient has vertical double vision, worse when reading or going down stairs. The most likely nerve:

- ☐ A. Trochlear nerve (CN IV)
- ☐ B. Facial nerve
- ☐ C. Optic nerve
- ☐ D. Abducens nerve

123. A patient cannot abduct the left eye, which turns inward. The lesion is:

- ☐ A. Right abducens nerve
- ☐ B. Left oculomotor nerve
- ☐ C. Left abducens nerve
- ☐ D. Left trochlear nerve

124. To test the superior oblique clinically, the patient looks:

- ☐ A. Straight lateral
- ☐ B. Up and out
- ☐ C. Up and in
- ☐ D. Down and in

125. A pituitary tumour compresses the optic chiasm from below. The visual defect:

- ☐ A. Left eye blindness
- ☐ B. Right homonymous hemianopia
- ☐ C. Bitemporal hemianopia
- ☐ D. Central scotoma only

126. Light shone into the right eye produces no constriction of either pupil; light in the left eye constricts both. The lesion is in the:

- ☐ A. Left oculomotor nerve
- ☐ B. Left optic nerve
- ☐ C. Right optic nerve
- ☐ D. Right oculomotor nerve

127. Light in either eye constricts the left pupil but never the right pupil. The lesion is:

- ☐ A. Right oculomotor nerve (efferent)
- ☐ B. Left oculomotor nerve
- ☐ C. Right optic nerve
- ☐ D. Left optic nerve

128. Touching the left cornea: neither eye blinks. Touching the right cornea: both eyes blink. The lesion is:

- ☐ A. Left facial nerve
- ☐ B. Right V1
- ☐ C. Left ophthalmic nerve (V1)
- ☐ D. Right facial nerve

129. Touching either cornea: the right eye blinks, the left eye never does. The lesion is:

- ☐ A. Left V1
- ☐ B. Right V1
- ☐ C. Left facial nerve
- ☐ D. Right facial nerve

130. A facial nerve lesion at the geniculate ganglion causes:

- ☐ A. Reduced tear secretion on that side
- ☐ B. Blindness
- ☐ C. Loss of corneal sensation
- ☐ D. Increased tear secretion on that side

131. A patient presents with sudden painless loss of vision in one eye and a cherry-red spot on fundoscopy. Most likely:

- ☐ A. Cataract
- ☐ B. Glaucoma
- ☐ C. Central retinal artery occlusion
- ☐ D. Conjunctivitis

132. After a punch to the eye, a patient has double vision on looking up and a numb cheek. Most likely:

- ☐ A. Lateral wall fracture
- ☐ B. Orbital floor blowout fracture
- ☐ C. Retinal detachment
- ☐ D. Optic canal fracture

133. A patient with headache and vomiting has bilateral swollen optic discs. This is:

- ☐ A. Uveitis
- ☐ B. Scleritis
- ☐ C. Cataract
- ☐ D. Papilloedema from raised intracranial pressure

134. A sudden painful red eye with halos and a mid-dilated pupil suggests:

- ☐ A. Myopia
- ☐ B. Acute closed-angle glaucoma
- ☐ C. Open-angle glaucoma
- ☐ D. Cataract

135. Yellow sclera in a patient with liver disease is called:

- ☐ A. Scleritis
- ☐ B. Uveitis
- ☐ C. Episcleritis
- ☐ D. Scleral icterus

136. A child's eye turns inward. The term is:

- ☐ A. Hypotropia
- ☐ B. Hypertropia
- ☐ C. Exotropia
- ☐ D. Esotropia

137. A patient reports flashes, floaters and a curtain over part of the vision. The emergency is:

- ☐ A. Chalazion
- ☐ B. Conjunctivitis
- ☐ C. Presbyopia
- ☐ D. Retinal detachment

138. A painless lump inside the eyelid from a blocked meibomian gland is a:

- ☐ A. Styne
- ☐ B. Chalazion
- ☐ C. Cataract
- ☐ D. Pterygium

139. A patient with mild ptosis, a small pupil and dry skin on one side of the face has a lesion of:

- ☐ A. The oculomotor nerve
- ☐ B. The trochlear nerve
- ☐ C. The cervical sympathetic chain (Horner)
- ☐ D. The facial nerve

140. A 45-year-old cannot read small print without holding it far away. The cause is:

- ☐ A. Retinal detachment
- ☐ B. Optic chiasm lesion
- ☐ C. Corneal abrasion
- ☐ D. Loss of lens elasticity (presbyopia)

Answer key

The single correct answer for each question, with a short explanation where it helps.

Eyeball I: coats, aqueous humour, refractive media, retina

1	C	
2	C	
3	A	The cornea is one sixth; the sclera is five sixths.
4	D	Maximum refraction occurs at the cornea.
5	B	
6	B	Yellow sclera from jaundice.
7	C	
8	A	Anterior uvea: iris and ciliary body; posterior: choroid.
9	C	It lies in front of the ora serrata and belongs to the uvea.
10	A	Mydriasis: darkness, sympathetic, dilator pupillae.
11	B	
12	D	
13	B	From the 2026 mock exam. Aqueous is almost protein-free; ascorbate and lactate are higher than in plasma (ascorbate is actively transported in).
14	B	Light absorption is the pigmented retinal layer.
15	B	Active secretion is relatively pressure independent.
16	A	
17	D	
18	D	
19	D	Prostaglandins actually enhance uveoscleral outflow.
20	A	
21	D	
22	C	
23	B	The lens is avascular.
24	A	The lens thickens (more convex).
25	A	
26	C	
27	D	
28	A	
29	A	
30	C	
31	C	
32	D	The fovea has the highest cone density.
33	A	The fovea is densely packed with cones.

34	B	
35	D	Cones contain iodopsin in three types (red, green, blue).
36	B	Photoreceptors and bipolar cells produce graded potentials.
37	D	From the 2026 mock exam.
38	C	

Eyeball II: extraocular muscles, retina and the visual pathway

39	B	
40	D	
41	D	
42	D	Superior rectus: elevation, adduction, intorsion.
43	A	Inferior rectus: depression, adduction, extorsion.
44	C	Superior oblique: depression, abduction, intorsion.
45	D	Inferior oblique: elevation, abduction, extorsion.
46	D	Superiors intort; inferiors extort.
47	D	Superiors intort; inferiors extort.
48	D	
49	A	
50	D	
51	B	
52	A	
53	A	
54	D	
55	B	
56	B	
57	D	
58	D	Brodmann area 17, around the calcarine sulcus.
59	A	
60	C	
61	B	Contralateral (right) homonymous hemianopia.
62	A	The temporal retina sees the nasal field.
63	A	The short ciliary nerves go to the sphincter (constrictor) pupillae.
64	B	From the 2026 mock: the pretectal nuclei project to ipsilateral and contralateral Edinger-Westphal nuclei.

Vascular supply and applied anatomy of the eye

65	A	
66	D	
67	B	Lacrimal is an orbital branch.
68	C	It enters the optic nerve about 13 mm behind the globe.

69	A	
70	C	
71	A	It passes through the supraorbital foramen.
72	B	
73	D	It also gives most of the anterior ciliary arteries.
74	B	
75	B	
76	C	Hypotropia is downward.
77	B	Most common is inward turning.
78	B	
79	B	
80	B	
81	D	Medial wall fractures accompany about half of floor fractures.
82	D	
83	B	
84	B	Orbicularis oculi closes the eye; its paralysis does not cause ptosis.
85	C	Corneal sensation is V1.
86	B	
87	C	
88	C	
89	A	
90	D	No cure, but early treatment can stop the damage.
91	A	

Motor and sensory innervation of the eye

92	C	Purely sensory: I, II, VIII.
93	D	Purely motor: III, IV, VI (and XI, XII).
94	C	Parasympathetic: III, VII, IX, X.
95	D	
96	B	
97	A	
98	C	The oculomotor nucleus is at the superior colliculus.
99	D	This is the lecture's answer. Note: the trochlear nerve crosses after leaving the nucleus, so textbooks say a nucleus lesion weakens the opposite superior oblique; a nerve lesion weakens the same side.
100	B	Interneurons drive the contralateral medial rectus for conjugate gaze.
101	C	
102	A	From the 2026 mock exam. The right eye turns medially.
103	B	Dioptric power increases.
104	A	
105	B	

106	B	
107	A	
108	B	
109	D	
110	D	The light reflex does not go to the visual cortex.
111	B	The effector is the sphincter pupillae.
112	C	
113	B	
114	D	Touching one cornea makes both eyes blink.
115	C	
116	A	
117	B	The synapse is in the pterygopalatine ganglion.
118	A	
119	C	
120	D	Its nucleus is in the midbrain and it enters the orbit through the superior orbital fissure.

Clinical cases (mixed)

121	C	Unopposed LR (VI) and SO (IV); levator and sphincter pupillae lost.
122	A	The superior oblique depresses the adducted eye.
123	C	Lateral rectus palsy → esotropia.
124	D	Adduct first, then depress.
125	C	From the 2026 mock exam.
126	C	Afferent defect on the right: the efferent pathways still work from the left eye.
127	A	Efferent defect on the right.
128	C	Afferent failure on the left.
129	C	Efferent failure on the left.
130	A	
131	C	The central retinal artery is an end artery.
132	B	Inferior rectus trapped; infraorbital nerve in the floor injured.
133	D	
134	B	
135	D	
136	D	
137	D	
138	B	
139	C	
140	D	